Serum vitamin B_{12} activity was determined with *Lactobacillus leichmannii* 313 (*ATCC* 7830). The method employed was the same as that described in Appendix No. 1, 1955, of the Swedish Pharmacopeia. Values less than 50 μμgm./ml. cannot be measured with sufficient accuracy with this method. Table 1 summarizes the results.

It will be seen that there is a pronounced difference in the serum vitamin B_{12} activity of patients with fish tapeworm anæmia and that of the control group. Seven out of eight anæmia patients gave concentrations less than 50 μμgm./ml. A somewhat higher value, 65 μμgm. vitamin B_{12} per ml. of serum, was found in this group in one case only. The control group displayed great variations ranging from 180 to 964 μμgm. vitamin B_{12} per ml., the average being 561 μμgm. There was no overlapping between the two groups.

To obtain the variations and the mean value of the vitamin B_{12} concentrations in the serum of fish tapeworm anæmia patients requires both more comprehensive material and a more accurate method. However, this introductory investigation shows that the vitamin B_{12} serum concentrations in fish tapeworm anæmia are of the same range as those in genuine pernicious anæmia. This supports the view concerning vitamin B_{12} deficiency in *D. latum* carriers with anæmia.

WOLMAR NYBERG

IV Medical Clinic,
University of Helsinki.

GUSTAF ÖSTLING

Pharmacological Laboratory,
Medica, Ltd.,
Helsinki.
June 28.

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Stem Canker (*Macrophomina phaseoli*) of Cotton Seedlings in the Sudan Gezira

IN some seasons cotton seedlings (*Gossypium barbadense* L.) in the irrigated Gezira area of the Sudan suffer considerable damage from stem canker caused by the fungus *Macrophomina phaseoli* (Maubl.) Ashby, also known as *Rhizoctonia bataticola* (mycelium) and *Sclerotium bataticola* (sclerotia).

First symptoms are the appearance of circular red spots up to about 5–6 mm. in diameter on the cotyledons of seedlings two or three weeks after emergence from the soil. Under favourable climatic conditions, notably high humidity, these spots spread and extend down the cotyledonary stalk to produce a conspicuous reddish-purple lesion at the cotyledonary node. These lesions become brown, dry and fibrous as dissolution of the parenchymatous cortical tissue of the young stem proceeds and eventually small black sclerotia develop. In severe cases the stem is completely girdled at the cotyledonary node, at which point affected plants tend to break (especially if handled), and thereafter die or suffer a severe check to growth. In warm dry weather cankered plants often recover and the disease dries out without causing much damage.

In the 1954–55 season, one of above-average post-sowing rainfall, stem canker was very prevalent in the central and southern Gezira and must have imposed a considerable check on early growth of the crop—a warning that this disease might become a major problem were earlier sowing of cotton introduced. With such earlier sowing the young crop would experience much heavier post-sowing rain than with present sowing in mid-August, by which date about two-thirds of the season's rain has normally fallen. Thus at the Gezira Research Farm in 1954 stem canker was severe and post-sowing rainfall was much heavier than usual—about 200 mm. as compared with long-term average of about 120 mm.

Although field observations strongly suggest that stem canker is favoured by excessively wet conditions in which the seedlings are exposed to prolonged water-logging, the method of infection is obscure. Attempts to produce the cotyledonary stage of the disease by spraying cultures of the pathogen and by other methods have so far failed, but characteristic stem lesions (from which the fungus was afterwards recovered) resulted from needle inoculation of the green cotyledonary node.

Pycnidia of *M. phaseoli* have not been observed on cotton in the Sudan and it is thought that seasonal infection originates from the soil, wherein the fungus survives the hot dry season in the form of small resistant sclerotia. Supporting this theory is the observation that soil fumigation with chloropicrin tended to increase stem canker incidence at the Gezira Research Farm in 1954–55. Plots fumigated with chloropicrin three to four weeks before sowing showed 24 per cent cankered seedlings (10 gallons/acre), 32 per cent (20 gallons/acre) and 45 per cent (40 gallons/acre) as compared with 24 per cent in unfumigated plots. Plots fumigated with 15 per cent methyl bromide showed 22·5 per cent (20 gallons/acre) and 23·5 per cent (40 gallons/acre), while 0·8 per cent formaldehyde drench using about 160 gallons of commercial formalin per acre gave 23·5 per cent cankered seedlings. The initial damage due to stem canker in the chloropicrin-treated plots was more than counterbalanced by the beneficial effects of fumigation, these plots producing more vigorous plants which finally yielded 32–48 per cent more seed cotton than the unfumigated plots, as compared with yield increases of 8–15 per cent for methyl bromide and 5 per cent for formalin. It is possible that the resistant sclerotia of *M. phaseoli* were able to withstand chloropicrin fumigation whereas soil micro-organisms lacking such resistant propagules were destroyed. Under the resultant conditions of reduced competition *M. phaseoli* flourished and incidence of stem canker increased, but this was not the case in fumigation with methyl bromide and formaldehyde.

S. A. J. TARR

Botany and Plant Pathology Section,
The Gezira Research Farm,
Wad Medani, Sudan.
July 28.

Relevance of Palæomagnetic Studies of Jurassic Rocks to Continental Drift

THE early palæomagnetic studies have largely been confined to the northern hemisphere, where wide sampling of the geological column in Great Britain and North America has done much to establish the reality of polar wandering. Runcorn[1] has also suggested, on the basis of a comparison of the British and American results, that a westward drift of some 24° would account for the systematic difference he has found in the results.

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Table 1

Region	Rocks sampled	Declination	Inclination	λ	K	N
N. Yorks*	Middle Lias —Lower Corallian. Arenaceous sediments	N. 3·4° E.	+66·9°	2·6°	87·5	36
Arizona	Kay nta Fm. (Lr. Jur.)	N. 6·9° W.	+43·5°	9·6°	50	6
North and South Rhodesia	Basalt lavas of Victoria Falls and Bulawayo	N. 30° W	−40·3°	4·6°	18·7	44
Tasmania	Intrusive dolerite sills	N. 35° W.	−85°	—	—	—

* Full details to be published.

The critical test of whether continental drift has occurred or not obviously lies in a study of the palæomagnetism of rocks of the same age in different continents. In this context the Jurassic period is of great importance, and it is now possible to assemble data from four continents as in Table 1 with the relevant statistics[2].

Although the arenaceous sediments of the British Jurassic are largely unsuitable for palæomagnetic work, being either too weak for measurement or unstable, there are sufficient which appear to be stable to give the results tabulated. Runcorn's rocks from Arizona fall in the same category, and his results, which are based on a few specimens, must be regarded with caution.

The Rhodesian basalt lavas were collected from the Bulawayo area and from the Victoria Falls, where fresh material was obtained from the second gorge with the aid of Dr. G. Bond. The basalts following above the Triassic Forest Sandstone are generally considered of lower Jurassic age. The position of the pole in the northern hemisphere (lying in Hudson Bay) is in latitude 61° N., longitude 76° W. Irving[3] carried out an extensive study of the intrusive dolerite sills of Tasmania. These sills, while certainly of Mesozoic age, have not been proved conclusively to be of Jurassic age. The ancient pole position, according to Irving, lay in latitude 50° N., longitude 23° W.

Thus it is clear that, while Britain in Jurassic times must be regarded as having approximately the same position as at present relative to the pole (the present pole lying within the circle of confidence of the Jurassic results), and the movement of America is not significant, it is equally clear that this is not the case in the southern hemisphere. The Rhodesian lavas indicate a movement of about 2,000 miles, and the Tasmanian results a movement of about 2,800 miles.

The inevitable conclusion is that continental drift must be accepted if the assumption of the coincidence of the axial dipole field with the rotational axis is accepted[4]. More exact dating of the time of drift and the relative positions of the continental blocks must depend on the more detailed work at present in hand in Great Britain, South Africa and Australia.

A. E. M. Nairn

Department of Physics,
King's College,
Newcastle upon Tyne.
July 3.

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A Statistical Information Theory of Visual Thresholds

In their recent letter on this subject Gregory and Cane[1] state that "perhaps other threshold and summation phenomena may be described in terms of statistical information available in neural information channels". A survey of available data on the relationship between the Weber–Fechner fraction $\Delta I/I$ and stimulus area A certainly supports this contention[2].

Gregory and Cane make the assumption that the number of independent observations n is proportional to the stimulus area A. Admittedly this is the simplest assumption; but there would seem to be good reasons for believing that the number of independent observations would more closely approximate the number of nerve fibres, f, operating in the area A, and f is certainly not proportional to A except perhaps for a limited range of areas in the fovea. If f is the correct estimate of the number of independent observations, $\Delta I/I$ should be proportional to $1/\sqrt{f}$. Using data on the variation of visual acuity across the retina to provide an estimate of f, $\Delta I/I$ is in fact proportional to $1/\sqrt{f}$ to a good approximation, over the wide range of stimulus areas from 10′ diameter to 40° diameter. There is some tendency for $\Delta I/I$ to be larger than predicted at very small field sizes, where eye-movements are probably a complicating factor, and at very large field sizes, where there is probably mental difficulty in utilizing simultaneously all the information supplied. If $\Delta I/I$ is plotted against $1/\sqrt{A}$ the results for field sizes up to about 2° are similar; but those for larger field sizes are markedly worse. Thus, for a field size of 36°, using $1/\sqrt{f}$ the predicted value of $\Delta I/I$ is about half the actual value, but using $1/\sqrt{A}$ the predicted value is about one-ninth of the actual value.

This type of statistical approach is very useful in tackling visual problems of this nature, and if certain assumptions are made it can lead to far-reaching conclusions. For example, if it is assumed that

$$N = f_1(c) \cdot f_2(I)$$

where N is the magnitude of the physiological response evoked by a stimulus of intensity I, and c is a measure of the state of adaptation, f_1 and f_2 being any functions, then by arguments given in the paper cited

$$N = N_0 + f_1(c)I^{r/k}$$

where N_0 is a constant, and r/k is the ratio of $\Delta N/N$ to $\Delta I/I$, $\Delta N/N$ being the physiological equivalent of the Weber–Fechner fraction. The above relationship implies a linearity between $\log (N - N_0)$ and $\log I$, and electrophysiological studies show that this is approximately true if N is the number of nerve impulses per second per nerve fibre.

I am grateful to Gregory and Cane for having directed my attention (in a private communication) to a slight error in the mathematics of my paper[2]; the conclusions of the paper are unaffected, but the matter will be discussed in a short note, which I hope to publish as a correction to the original paper.

R. W. G. Hunt

Research Laboratories,
Kodak Limited, Wealdstone, Harrow,
Middlesex. July 20.

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